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DEPARTMENT OF CSE/CS&IT

PROJECT ABSTRACT SUBMISSION

COURSE: Big Data Analytics (23DSB3303R)

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TITLE OF THE PROJECT	Smart Customer Purchase Analytics & Recommendation System	
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Abstract

The **Smart Customer Purchase Analytics & Recommendation System** is a data analytics project that focuses on understanding customer purchase behaviour using big data analytics techniques. In today's digital retail environment, organizations generate very large amounts of transaction data. This data contains useful information about customer preferences, buying patterns, and product performance. However, without proper analysis, this data cannot be effectively used for business decision-making.

This project uses historical customer purchase data to perform data cleaning, data integration, and exploratory data analysis. Different analytical techniques are applied to study sales trends, customer segmentation, product-wise performance, and purchasing frequency. Data visualization tools are also used to present the results clearly through graphs and charts such as sales trends, category-wise revenue, and customer spending patterns.

In addition, a simple rule-based recommendation method is implemented to suggest products based on customers' past purchases and frequently bought items. The system helps in identifying high-value customers, popular products, and possible cross-selling opportunities. Overall, this project shows how big data analytics and data visualization support data-driven business decisions and act as a basic foundation for intelligent recommendation systems in real-world retail applications